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Mini-Project (22CSMP607) Synopsis On:

"XIPER - SECURE DECENTRALIZED CHAT APP"

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ABSTRACT

This project presents Xipher, a secure and decentralized mobile messaging application designed to uphold user privacy, resist surveillance, and operate without centralized control. Built on top of the open-source Session framework, Xipher inherits a robust foundation of privacy-focused architecture while introducing enhancements tailored for real-world deployment and security-conscious users.

Xipher leverages the Oxen Service Node Network to ensure metadata-free, anonymous messaging without the need for phone numbers, email IDs, or centralized servers. All communication is protected through end-to-end encryption, ensuring that no third party — including service providers — can access the content or metadata of messages.

Developed using Android Studio, the app features a custom local authentication system using PIN or biometric lock, providing an added layer of device-level protection. By focusing on clean interface design and removing non-essential components, Xipher offers a lightweight and distraction-free messaging experience optimized for privacy.

Additionally, the project demonstrates key cybersecurity and software engineering principles such as open-source collaboration, decentralized architecture, cryptographic security, and privacy by design. It emphasizes practical development practices for secure mobile communication in a world where digital privacy is increasingly threatened.

Xipher stands not only as a technical achievement, but also as a statement — that user privacy is a right, not a feature. The application is designed to be fully deployable on the Google Play Store, showcasing the potential of secure open-source tools to combat modern digital threats.

OBJECTIVES

The main objective of this project is to develop Xipher, a secure mobile messaging application that emphasizes user privacy, anonymity, and protection against digital surveillance.

The project aims to achieve the following:

- To create a messaging platform that does not rely on phone numbers or personal identifiers, allowing users to communicate anonymously.
- To implement strong security features, including end-to-end encryption and local app-lock options using PIN or biometric authentication.
- To ensure smooth and efficient messaging with a simple and user-friendly interface suitable for real-world use.
- To apply key cybersecurity principles in the design and development of a secure communication system.
- To publish the application on the Google Play Store for public access and demonstrate its practical usage.

This project showcases how privacy-focused messaging systems can be built using secure communication techniques and responsible development practices.

METHODOLOGY

Project Initialization:

The development of Xipher began by building upon a well-established, open-source secure messaging framework. This provided a strong foundation for implementing decentralized communication while allowing further customization based on our objectives.

Anonymous Identity Generation:

Xipher allows users to create accounts without phone numbers or emails. Instead, a randomly generated Session ID serves as the user's identity, promoting anonymity and metadata-free communication.

Secure Messaging Architecture:

Leveraging the power of a decentralized service node network, Xipher ensures end-to-end encrypted messaging without storing IP addresses or message metadata. This decentralized architecture helps protect against surveillance and tracking.

Local App Lock Integration:

To add an additional layer of security, a custom app lock mechanism was implemented using biometric authentication or PIN, preventing unauthorized access even if the device is compromised.

User Interface and Experience Enhancements:

The app interface was streamlined to provide a clean, minimal, and distraction-free messaging experience. Navigation, performance, and aesthetics were prioritized to enhance real-world usability.

Core Feature Optimization:

Unnecessary modules from the open-source base were removed to reduce app size and improve performance. Essential messaging functionalities were retained and tested for seamless operation.

Testing and Debugging:

The application was tested across different Android versions and emulators to ensure compatibility and functionality. Errors were debugged, and iterative improvements were made during development.

Deployment Preparation:

Final builds were optimized and prepared for deployment on the Google Play Store. The app was signed, versioned, and packaged according to official publishing standards.